DR. MUNDO · SCOPE V2

The Data

Where every number comes from, how the tables fit together, and the reasoning behind each decision.

Makati Medical Center · PhilHealth Annex B · built and verified August 2026

The one rule

No number exists unless it was published. If a price isn't in an official source, there is no row for it — and the app says so rather than estimating. Every figure in the database traces to either a Makati Medical Center catalogue item code or a PhilHealth Annex B case rate. **Nothing is synthetic.**

This is the claim worth defending in the presentation. It is also why several things are deliberately *missing* from the dataset — each gap is a choice to be honest rather than to fill space with a plausible-looking figure.

The dataset at a glance

6,167 items extracted from MMC	**2,446** priced rows kept	**4,312** PhilHealth case rates
7 database tables	**51** procedures mapped to PhilHealth	**0** invented numbers

What this document covers

1. **Where the data came from** — the two sources, and the trap that nearly cost us the whole dataset
2. **How it was extracted** — and two bugs that would have shipped silently
3. **The ERD** — seven tables and why each one is separate
4. **Where the data sits in the flow** — from question to answer
5. **The crosswalk** — the hardest part, done by hand, and why
6. **Two safeguards** — the flags that stop the app lying politely
7. **Known gaps** — stated plainly, because they are the point

1 Where the data came from

Two sources. One national and public, one hospital-specific and easy to miss.

PhilHealth Annex B

4,312 procedure case rates. The national list of what PhilHealth pays per procedure, keyed by RVS code. Published as a PDF; scraped once into CSV.

Verified current: PhilHealth Circular 2024-0037 applied a 50% uplift to select case rates. We checked — **98.4%** of the 4,312 rates divide cleanly by 1.5 into whole pesos (appendectomy = ₱31,200 × 1.5 = ₱46,800). The data already includes the adjustment.

MakatiMed price list

6,167 catalogue items. The hospital's own published prices — every item with a low price, a high price, a department, an official item code, and an "as of" date.

This is what makes the project defensible: these are not estimates or market averages. They are the hospital's own published figures, and anyone can look up any item code and check.

The trap. Fetching `makatimed.net.ph/price-list` returns a blank page, and it is easy to conclude the hospital publishes nothing — which is what most large Philippine private hospitals actually do. St. Luke's and Asian Hospital genuinely publish no itemised prices. But MakatiMed's page is a **searchable database that renders empty until you search.** We nearly abandoned Makati Med and switched hospitals over this.

Why this matters for the business case

The project's premise is that hospital prices are opaque. Confirming by direct search that the country's major private hospitals publish nothing itemised — and that the one exception hides its data behind a search box — is **evidence for the thesis**, not just a research inconvenience. It is worth saying out loud in the presentation.

What a raw extracted row looks like

```
# one item, exactly as published by Makati Medical Center
mmc_code   : 04360316
name       : LIPID PROFILE (HDL LDL CHOL TRIG) SERUM
mmc_type   : LAB-CHEMISTRY          ← department, not a category
price_low  : 4,450
price_high : 8,700
as_of      : March 10, 2026         ← varies per item
```

Note: *`mmc_type` is the hospital's internal department, not a clinical category. "DELIVERY SURGERY" contains real operations, a blood test, and four room-rental charges. Classification therefore had to be done per item, not per department.*

2 How it was extracted

Figure 1: *The extraction pipeline. A one-time offline build step — the app itself never contacts the internet when answering a question.*

The search is a plain form POST, and crucially the **prices are already inside the returned HTML**, embedded in a per-row script that builds the "See Price" pop-up. So one request returns fully-priced rows; no clicking through 6,167 modals.

Two bugs that would have shipped silently

Bug 1 — the wrong vocabulary. The first sweep searched by modality: "X-RAY", "ULTRASOUND", "CT SCAN". But MakatiMed names imaging by *body part* — `CHEST PA`, `WHOLE ABDOMEN`, `KIDNEYS COLOR DOPPLER`. The result looked impressive (1,204 items) and was missing chest X-ray, ECG, abdominal ultrasound and HbA1c — **the most common items on a real doctor's request.**

Fix: stop guessing their vocabulary. The search is unbounded — a single letter "A" returns 5,195 rows — so the catalogue can be enumerated exhaustively instead.

Bug 2 — vowels. The enumeration first swept vowels only (A, E, I, O, U). But `HDL` and `LDL` **contain no vowels**, so the method structurally could not see them. Those two tests are members of the lipid panel — the centrepiece of the panel-vs-individual feature.

Fix: sweep the full alphabet. Every catalogue name contains at least one letter, so coverage is now provably complete.

How Bug 2 was caught. Not by review — by a guard written into the build. The panel table refuses to emit a panel unless *every* member test is priced. Rather than silently publishing a half-grounded comparison, it produced zero rows and named the missing members. **The dataset told us it was wrong.** This is worth mentioning if anyone asks how you know the data is trustworthy.

3 The database — entity relationship diagram

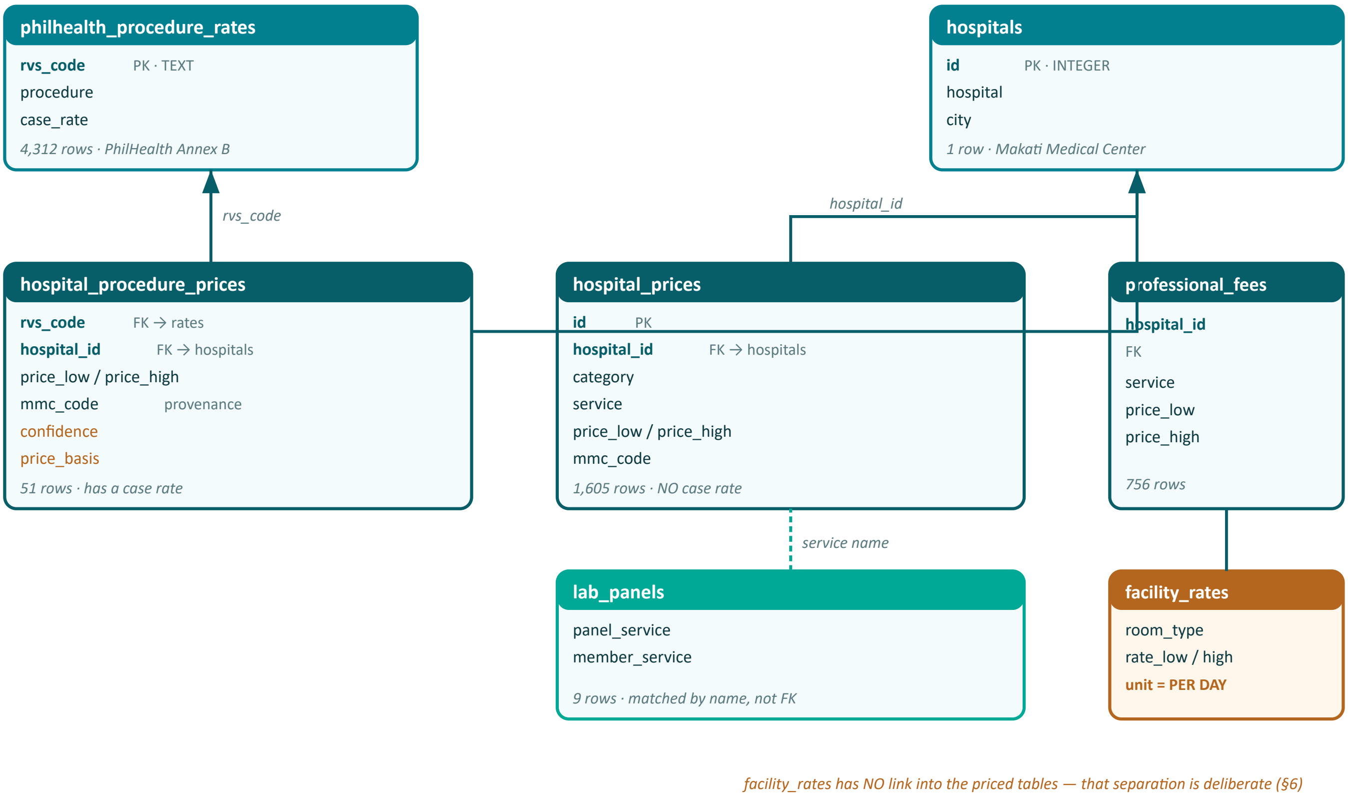

Figure 2: *Seven tables. Arrows point from the many side to the one side. Solid = enforced foreign key. Dashed = matched by service name, because a service name is not unique.*

Why the split between the two price tables

This is the design decision most worth understanding. `hospital_procedure_prices` holds only procedures we could tie to a PhilHealth RVS code — so a case rate can be subtracted. `hospital_prices` holds everything else: lab tests, imaging, and the 19 procedures whose RVS code could not be honestly determined. **The table an item lands in decides whether coverage math is even possible.**

Table	Rows	Contents
hospital_prices	1,605	842 laboratory · 566 imaging · 178 diagnostic · 19 procedures without a case rate
professional_fees	756	Surgeon and anaesthesiologist, itemised by specialty
hospital_procedure_prices	51	Procedures with a verified RVS code
facility_rates	34	Room & board, per day
lab_panels	9	Panel → member test mappings

4 Where the data sits in the flow

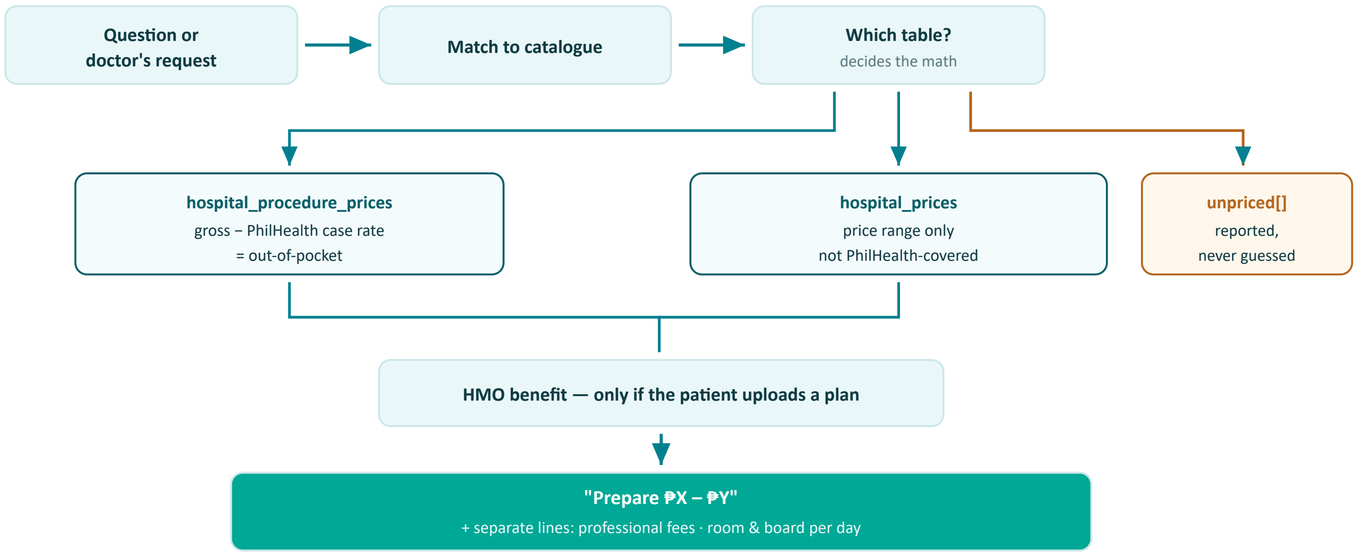

Figure 3: *The data's role in answering. The branch at the top is the whole design: which table an item lives in determines whether PhilHealth math applies at all.*

The conservation rule. Every item read from a request must end up in exactly one bucket: `priced + unpriced + needs_confirmation = extracted`. Silently dropping an item would understate someone's hospital bill and would be *invisible* in the output. This is enforced as a hard failure, not a warning.

Retrieval — how a Taglish phrase reaches a clinical name

MakatiMed's names are long and clinical. Nobody types `LIPID PROFILE (HDL LDL CHOL TRIG) SERUM`. Two layers work together: semantic embeddings over all 1,602 service names, plus a curated alias list that boosts common lay and Taglish phrasings. Verified working:

What the user types	What it resolves to	Score
magkano ang cbc	CBC (COMPLETE BLOOD COUNT)	1.49
chest xray	CHEST PA	1.55
ultrasound ng tiyan	WHOLE ABDOMEN	1.23
tanggal apdo	LAPAROSCOPY, SURGICAL; CHOLECYSTECTOMY	1.09
magkano ang tuli	CIRCUMCISION, USING CLAMP OR OTHER DEVICE	1.07

5 The crosswalk — the hardest part

MakatiMed uses its own item codes. PhilHealth pays by RVS code. **Nobody publishes a mapping between them.**

Without this link there is no coverage math at all — you have a hospital price and a national case-rate list with no way to connect them. Building it was the single largest piece of judgement in the dataset.

Why it could not be automated

An automatic matcher was built first. It scored candidates on surgical word endings (–ECTOMY, –OSTOMY, –PLASTY). That works for `LAPAROSCOPIC CHOLECYSTECTOMY` and collapses entirely on acronyms — CABG, TAHBSO, FESS, ACL, VATS have no such word. With nothing to grip on, it fell back to generic word overlap and produced confident nonsense:

MakatiMed item	It proposed	Which is actually…
CESAREAN SECTION MOTHER	88331	Pathology consultation with frozen section
CABG 3 VESSELS	11403	Excision of a benign skin lesion
Endoscopic sinus surgery	11770	Excision of a pilonidal cyst
USE OF WATERPROOF DOPPLER	93556	Not a procedure at all — rated **"high confidence"**

Why this is the dangerous kind of error. A wrong code still returns a real number from a real table. The answer looks perfectly reasonable and is tens of thousands of pesos wrong. Cholecystectomy alone has three case rates — ₱60,450 / ₱90,675 / ₱104,130 — so picking the wrong variant is a ₱43,680 error that nothing downstream would catch.

How the 51 mappings were actually made

By hand: expand the acronym into the clinical operation, read the candidate RVS descriptions, and confirm the operation, the surgical approach and the extent all agree. PhilHealth codes laparoscopic and open operations separately, so approach must match. A worked example:

Candidate for "LAPAROSCOPIC CHOLECYSTECTOMY PACKAGE"	Case rate	Verdict
47562 — Laparoscopy, surgical; cholecystectomy	**₱60,450**	**Chosen**
47563 — …with cholangiography	₱60,450	Bundles work MMC never mentions
47564 — …with common-duct exploration	₱90,675	₱30k richer — wrong operation
47600 — Cholecystectomy	₱60,450	The *open* operation

The tie-breaking rule

Where two codes are both defensible, take the **lower** case rate. Less assumed PhilHealth help means a *higher* estimated out-of-pocket, so an uncertain mapping makes the patient over-prepare rather than under-quote. **Over-preparing is survivable. Under-quoting a hospital bill is not.**

41	**9**	**19**
high confidence	medium confidence	honestly unmapped

Every mapping records its reasoning and the alternatives it rejected. The 19 unmapped keep their real price and report PhilHealth coverage as **undetermined** *— not zero.*

6 Two safeguards built into the data

Both exist to stop the app saying something reasonable-sounding and false.

Safeguard 1 — `price_basis`

A PhilHealth case rate is **all-in**: it covers the facility fee and the doctor's fee together. So if a case rate is *larger* than the most MakatiMed charges, MakatiMed cannot be billing the whole episode — it is billing a component. Four rows behave this way:

Item	MMC charges	PhilHealth pays	Flag
Colposcopy	₱4,100	₱15,639	component
LEEP	₱4,860	₱18,915	component
IUD insertion	₱2,060	₱3,900	component
Vaginal hysterectomy	₱40,590	₱59,085	component

Without this flag the app concludes **"fully covered"** from a partial price — and the patient turns up expecting to pay nothing. The flag makes that impossible.

A subtlety worth knowing. The comparison is against the *ceiling* price, not the floor. A case rate that merely sits *inside* a wide range is ordinary partial coverage — lithotripsy is ₱22,700–70,900 against a ₱35,100 case rate, which is normal. An earlier version compared against the floor and wrongly flagged it. Only a case rate above the ceiling proves a partial price.

Safeguard 2 — facility rates cannot be summed

Room and board is real, published, and **charged per day**: ward ₱1,810, ICU ₱9,900, private ₱10,900, suite ₱21,800 per day. To add it to a bill estimate you would need the length of stay — which a doctor's request never states.

Assuming one would invent the single largest number on the page: a 3-day versus 7-day suite stay is an **₱87,000 swing**, a fabricated multiplier sitting on top of carefully sourced figures. So facility rates live in **their own table**, with no link into the priced tables. The separation is structural — not a convention someone can forget.

How it is shown instead: *"If admitted, room & board is billed separately: ₱1,810/day (ward) to ₱21,800/day (suite)."* Real numbers, useful for planning, nothing invented.

Why the gap is narrower than it looks

- MakatiMed's operating-room prices are **all-in packages** — theatre time and room use are already inside them.
- PhilHealth case rates **already bundle a facility fee**, so adding room charges *and* subtracting a full case rate would double-count in the patient's favour — the more harmful direction to be wrong in.

7 Known gaps — and why they are the point

These are the strongest evidence that the "no invented numbers" rule was actually enforced, including where it was inconvenient.

Appendectomy has no hospital package price. MakatiMed publishes only professional fees for it — surgeon ₱75,000–105,000 open, ₱100,000–140,000 laparoscopic — and no operating-room package.

This was the midterm's flagship demo, so **the demo needs changing.** The app can still answer usefully (case rate ₱46,800 plus real surgeon fees) but must state that the hospital portion is missing. It would have been easy to invent a package price here. We didn't.

Gap	Why
19 procedures with no case rate	No honest RVS match exists — e.g. "VATS" names the approach, not the operation; 16 candidate codes span ₱23,634–90,675 and nothing selects among them
No cataract package mapping	The PhilHealth catalogue has no primary cataract-extraction-with-lens code
No fecalysis	Not in MakatiMed's catalogue under any name we could find
No HMO table	By design — the patient uploads their own plan, so nothing synthetic is committed
Consultations not extracted	Outside the three agreed scope categories

Why this section matters most. Each gap above is a place where inventing a plausible number would have been easy and nobody would have noticed. The dataset is trustworthy precisely because these are empty.

What the data delivers

All three agreed scope categories, every figure from a published price.

Scope category	Coverage	Source
(a) Procedures	51 + 19	MMC operating-room and delivery packages; 51 linked to PhilHealth
(b) Radiologic studies	566	X-ray, ultrasound, CT, MRI, nuclear medicine, mammography
(c) Blood tests — individual *and* panels	842	Chemistry, haematology, immunology, microbiology, plus panel mappings

Scope (c) proved on real numbers

The panel-versus-individual claim — that a bundled panel is cheaper than buying its components separately — holds across all three lipid panels:

Panel	As a panel	Bought separately	You save
HDL · LDL · Cholesterol · Triglycerides	₱4,450–8,700	₱4,520–9,400	₱70–700
HDL · LDL · Triglycerides	₱3,750–6,700	₱3,810–8,100	₱60–1,400
HDL · LDL	₱2,400–4,400	₱2,550–5,400	₱150–1,000

Every figure above is a published MakatiMed price — the strongest single demo in the dataset, and a real insight a patient could act on.

The one-sentence version. Six thousand published hospital prices and four thousand PhilHealth case rates, joined by a hand-verified crosswalk, with every figure traceable to an official item code — and where the source was silent, the dataset stays silent too.